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United States Patent	12409518
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Berney; Chandler

Fastener removal tool

Abstract

A powered tool for removing mechanical fasteners from materials. The fastener removal tool includes: a fastener puller for engaging the head of a fastener; a stand-off mount for placing against a surface having a fastener therein; a rod joined to the fastener puller; and a power source operatively connected to the rod. The fastener removal tool may exert a pulling force on the head of a fastener in the form of repeating intermittent strikes on the underside of the head of a fastener with said fastener puller. The fastener removal tool may have an indicator mechanism for indicating whether the fastener puller completely engages a fastener. The fastener removal tool may have a fastener lubricating system for lubricating difficult to remove fasteners.

Inventors: Berney; Chandler (Oklahoma City, OK)

Applicant: Government of the United States, as represented by the Secretary of the Air Force (Wright-Patterson AFB, OH)

Family ID: 88416737

Assignee: United States of America as represented by the Secretary of the Air Force (Wright-Patterson AFB, OH)

Appl. No.: 17/958502

Filed: October 03, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230339054 A1	Oct. 26, 2023

Related U.S. Application Data

us-provisional-application US 63334954 20220426

Publication Classification

Int. Cl.: B23P19/00 (20060101); B23P19/06 (20060101); B25B27/02 (20060101)

U.S. Cl.:

CPC B23P19/06 (20130101); B25B27/026 (20130101);

Field of Classification Search

CPC: B23P (19/00); B23P (19/06); B23P (19/027); B23P (19/043); B23P (19/066); B23P (19/10); B25B (27/00); B25B (27/06)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2706103	12/1954	Stambaugh	254/19	B25C 11/02
2735649	12/1955	Swallert	254/18	B25C 11/02
2902751	12/1958	Sommer	N/A	N/A
3069761	12/1961	Sommer	294/206	B25B 27/026
3643918	12/1971	Ellis	254/19	B25C 11/02
6113073	12/1999	Lefavour	254/24	E01B 29/26
6526641	12/2002	Latham	29/239	E21C 35/18
6601277	12/2002	Swanson	29/259	B25B 27/023
7963012	12/2010	Brasher	29/244	A63B 60/06
9457458	12/2015	Wagstaff et al.	N/A	N/A
9616554	12/2016	Hendrix, II	N/A	N/A
10688637	12/2019	Hare et al.	N/A	N/A
11123848	12/2020	Wojciechowski et al.	N/A	N/A
2004/0187283	12/2003	Oser	29/252	B25B 27/0035
2012/0317767	12/2011	Bathurst	29/243.53	B23P 19/00
2016/0023334	12/2015	Rowe	29/244	F16D 3/405
2023/0339054	12/2022	Berney	N/A	B25B 27/18

Primary Examiner: Wilson; Lee D

Attorney, Agent or Firm: AFMCLO/JAZ

Background/Summary

(1) Pursuant to 37 C.F.R. § 1.78(a)(4), this application claims the benefit of and priority to prior filed Provisional Application Ser. No. 63/334,954, filed Apr. 26, 2022, which is expressly incorporated herein by reference.

FIELD OF THE INVENTION

(1) The present invention relates generally to tools for removing fasteners and, more particularly, to a powered tool for removing fasteners.

BACKGROUND OF THE INVENTION

(2) Fasteners for fastening various components together are well known. In some cases, fasteners may need to be removed. Fasteners may, for example, become stuck in a component and be partially exposed so that they stick out of the surface of a component.

(3) TAPER-LOK® fasteners used on aircraft structures which need to be removed often become stuck in their housing with the neck of the fastener partially exposed. The typical method of removing these fasteners is by using a rivet gun to drive them out from below and wedges to simultaneously apply upward pressure. This method is dangerous, time consuming, and inefficient.

(4) Methods to remove other stuck fasteners include the use of tools such as pry bars, pliers, hammers, and wedge tools in order push, pry, or pull fasteners out using manual force. These methods are inefficient and slow, limited to the force the person using these tools can apply, potentially damaging to the surface the fasteners are connected to, potentially dangerous to the person performing the task, and unwieldy and bulky.

(5) The search for improved tools and methods for removing fasteners has, therefore, continued.

SUMMARY OF THE INVENTION

(6) The present invention relates generally to tools for removing fasteners and, more particularly, to a powered tool for removing fasteners. The powered tool can be pneumatic, hydraulic, magnetic, electric, or electronic.

(7) While the invention will be described in connection with certain embodiments, it will be understood that the invention is not limited to these embodiments. To the contrary, this invention includes all alternatives, modifications, and equivalents as may be included within the spirit and scope of the present invention.

(8) According to one embodiment, a fastener removal tool is provided for removing a fastener having a head that is partially protruding from the surface of an article. The fastener removal tool comprises: a fastener puller for engaging the head of a fastener, the head of the fastener having a top surface and an underside; a stand-off mount for placing against a surface having a fastener therein, the stand-off mount having two ends comprising a surface-contacting end and an opposing end, a passageway between the ends; a first rod joined to the fastener puller and passing part of the way through the passageway in the stand-off mount; and a power source having a piston rod that is joined to said first rod, wherein the power source controls the motion of the first rod, wherein the fastener removal tool is configured to have one setting in which it exerts a pulling force on the head of a fastener in the form of repeating intermittent strikes on the underside of the head of a fastener with the fastener puller.

(9) According to another embodiment, a method of removing a fastener from an article is provided. The fastener has a body and a head joined to the body. The fastener head has a top surface and an underside. The fastener head protrudes from the surface of the article so that a gap is provided between the underside of the fastener head and the surface of the article. The method comprises: a) providing a powered fastener removal tool comprising: a fastener puller for engaging the head of a fastener; a stand-off mount for placing against a surface of an article having a fastener therein, wherein the stand-off mount has a passageway therethrough and the fastener puller is positioned for reciprocating motion in the passageway; and a power source operatively connected to the fastener puller, wherein the power source is in communication with an actuator that controls the motion of the fastener puller; b) placing the stand-off mount against the surface of the article and engaging the head of the fastener with the fastener puller; c) actuating the fastener puller with the actuator; and d) exerting a repeating intermittent striking force on the underside of the head of a fastener with the fastener puller using the powered fastener removal tool.

(10) Additional objects, advantages, and novel features of the invention will be set forth in part in the description which follows, and in part will become apparent to those skilled in the art upon examination of the following or may be learned by practice of the invention. The objects and

advantages of the invention may be realized and attained by means of the instrumentalities and combinations particularly pointed out in the appended claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate embodiments of the present invention and, together with a general description of the invention given above, and the detailed description of the embodiments given below, serve to explain the principles of the present invention.
- (2) FIG. 1 is a side view of one embodiment of a fastener removal tool.
- (3) FIG. 1A is an enlarged perspective view of one embodiment of the fastener puller that can be used with the fastener removal tool.
- (4) FIG. 2 is an exploded view of the fastener removal tool.
- (5) FIG. 3 is a side view of the fastener removal tool in FIG. 1, shown from the other side, with a portion partially cut away.
- (6) FIG. 4 is an exploded view of the handle plate assembly.
- (7) FIG. 5 is a perspective view of an alternative embodiment of the fastener removal tool at includes additional features.
- (8) FIG. 6 is a perspective view of the alternative fastener removal tool in FIG. 5 shown from another angle.
- (9) FIG. 7 is a partially cut away side view of a portion of the fastener removal tool in FIG. 5.
- (10) FIG. 7A is a cross-section taken along line 7A-7A of FIG. 7.
- (11) FIG. 8 is a bottom perspective view showing one embodiment of a sensor inside the fastener puller.
- (12) FIG. 9 is a cross-sectional view of a portion of the fastener removal tool showing the fastener puller initially engaging the head of a fastener.
- (13) FIG. 10 is a cross-sectional view similar to that shown in FIG. 9 showing a sequence of positions of the fastener puller after it has partially pulled the fastener out of the surface to which it was affixed.
- (14) FIG. 11 is a side view of an embodiment of a fastener removal tool with a selector switch for selecting different modes of applying a force to remove a fastener.
- (15) It should be understood that the appended drawings are not necessarily to scale, presenting a somewhat simplified representation of various features illustrative of the basic principles of the invention. The specific design features of the sequence of operations as disclosed herein, including, for example, specific dimensions, orientations, locations, and shapes of various illustrated components, will be determined in part by the particular intended application and use environment. Certain features of the illustrated embodiments have been enlarged or distorted relative to others to facilitate visualization and clear understanding. In particular, thin features may be thickened, for example, for clarity or illustration.

DETAILED DESCRIPTION OF THE INVENTION

- (16) The present invention relates generally to tools for removing fasteners and, more particularly, to a powered tool for removing fasteners. The powered tool can be powered by pneumatic, hydraulic, magnetic, electric, or electronic force. The application of pneumatic and hydraulic forces may be referred to generically herein as “fluid powered”.
- (17) The fastener removal tool and the methods described herein can be used to remove many different types of fasteners in many different industries. The types of fasteners and industries the fastener removal tool and the methods can be used in include, but are not limited to: aircraft maintenance/construction, automotive maintenance/construction, aquatic

maintenance/construction, carpentry, general industrial solutions, general construction uses, general maintenance uses, general disassembly and demolition uses, ground stake removal, post removal, door/window removal.

(18) FIG. 1 shows one non-limiting embodiment of a fastener removal tool **20**. This particular fastener removal tool may be particularly useful in removing TAPER-LOK® tapered shank bolt fasteners. TAPER-LOK® fasteners are commonly used to fasten aircraft splice plates to portions of aircraft. TAPER-LOK® fasteners have a head and a tapered body or shank which is wider near the head than at the end. The shank is threaded at the end for receiving a nut. These fasteners are installed into a tapered hole, and are designed to provide an interference fit upon installation. The fastener removal tool **20** can, however, be used to remove any of the types of fasteners described herein.

(19) The fastener removal tool **20** when in use generally comprises: a fastener head puller (“fastener puller” or “fastener gripper”) **22** for engaging the head of a fastener; a stand-off mount **24** for placing against a surface having a fastener therein; a rod **26** (shown in FIG. 2) joined to the fastener puller **22** and passing part of the way through a passageway in the stand-off mount **24**; a power source such as a pneumatic (or air) cylinder **30** joined to the rod **26**. The fastener removal tool **20** may have a longitudinal axis, L. In the case of a pneumatically powered fastener removal tool **20**, the pneumatic cylinder **30** controls the motion of the rod **26**. An air flow direction control may be operationally connected to the pneumatic cylinder **30**. The fastener removal tool **20** may also comprise a handle **34**. The fastener removal tool **20** may comprise additional components as described herein.

(20) FIG. 1A shows one embodiment of a fastener puller **22**. The fastener puller **22** may comprise any type of component that is suitable for engaging the head of a fastener. The fastener puller **22** conforms to the head and neck of the targeted fastener in order to grasp the fastener which prevents or reduces damage to the fastener and work surface. The fastener puller **22** should also be capable of withstanding the application of a force thereto, which is in turn, exerted on the head of the fastener, in order to remove fasteners from various articles. The fastener puller **22** may have different configurations for different sizes and types of fasteners. The fastener puller **22** is used as part of the fastener removal tool **20** during use. However, the fastener removal tool **20** and the fastener puller **22** may be provided separately (much like a drill and drill bits) so that it need not comprise part of the fastener removal tool **20** per se.

(21) In the embodiment shown in FIGS. 1A to 3, the fastener puller **22** has a generally cylindrical configuration with a longitudinal axis L1. The fastener puller **22** has an inner end **22A** and an outer end **22B**. FIG. 3 shows that the fastener puller **22** is joined adjacent its inner end **22A** to the rod **26** of the fastener removal tool **20**. More specifically, in the embodiment shown, the inner end **22A** of the fastener puller **22** is joined to the rod **26** of the fastener removal tool **20** by providing an axial bore **22C** that extends from the inner end **22A** of the fastener puller part of the way into the interior of the fastener puller **22**. The axial bore **22C** may be threaded or smooth (unthreaded). The rod **26** fits into this axial bore **22C**. The rod **26** can be retained in the bore **22C** by a retaining element such as a set screw **36** that passes through an opening **22D** in the side of the fastener puller **22** that is oriented perpendicular to the axis A1 of the fastener puller **22**.

(22) The outer end **22B** of the fastener puller **22** is used to engage a fastener. As shown in FIGS. 1A to 3, the outer end **22B** of the fastener puller **22** has a recessed pocket **38** formed therein that is configured to fit the target-sized fastener. The recessed pocket **38** has an opening that is configured to accept the head of a fastener, and an engagement feature **40** that fits under the head of a fastener. The recessed pocket **38** includes a channel for the neck of the fastener, the sides of which channel are defined by the engagement feature **40**. The fastener puller **22** can be made of any suitable material by any suitable process. In some cases, the fastener puller **22** is machined from a 304 stainless steel rod.

(23) The stand-off mount **24** may have a generally hollow cylindrical configuration. The stand-off

mount has two ends comprising an inner end **24A** and an opposing surface-contacting end **24B**. The surface-contacting end **24B** can be tapered inward toward the longitudinal axis **L**, and can then widen into a flange **24C** at the distal end thereof. The flange **24C** will rest against the surface containing the fastener that will be removed. A passageway **44** (shown in FIG. **3**) extends between the ends **24A** and **24B**. The sides of the stand-off mount **24** may have at least one recess **24D** therein so that the user will be able to see the fastener. The stand-off mount **24** may have multiple openings or recesses **24D** formed on the surface-contacting end **24B** that allow for viewing of the work area and increased maneuverability. If there are multiple openings, these may be formed between multiple distinct contact surfaces or “feet” that contact the work surface. The stand-off mount **24** can be made of any suitable material by any suitable process. Suitable materials include, but are not limited to metals, plastics, and composite materials such as carbon fiber nylon. Suitable processes for making the stand-off mount **24** include, but are not limited to 3D printing and casting.

(24) The stand-off mount **24** may be joined to a stand-off mount-receiving component (or face plate) **46** as shown in FIG. **2**. The stand-off mount-receiving component **46** may comprise a plate **48** having an opening **50** in its center. A receiving element, which may be a cylindrical element **52**, is joined to the plate **48** and extends outward perpendicularly from one side of the plate **48** and surrounds the opening **50** in the plate. The plate **48** and the cylindrical element **52** can comprise two components that are joined together, or they may comprise a single integrally-formed component. In the embodiment shown in the drawings, the plate **48** has a plurality of holes **54** therein for joining the stand-off mount-receiving component **46** to an air cylinder **30** using screws **56**. FIG. **3** shows that the cylindrical element **52** may have at least one opening in the side thereof for a fastener (such as a set screw **36**) to pass through in order to join the stand-off mount **24** to the stand-off mount-receiving component **46**. The stand-off mount-receiving component **46** can be made of any suitable material by any suitable process including those described for the stand-off mount. In one case, the stand-off mount-receiving component **46** may be a 3D printed carbon fiber nylon structure.

(25) The rod (“threaded rod” or “first rod”) **26** joins the fastener puller **22** to the piston rod of the air cylinder. The rod **26** has an inner end **26A** and an outer end **26B**. The rod **26** extends part of the way into the passageway **44** in the stand-off mount **24**. The inner end **26A** of the rod **26** is joined to the actuator of the air cylinder. The inner end **26A** of the rod **26** may be joined to the actuator of the air cylinder in any suitable manner. For example, at least a portion of the rod **26** at inner end **26A** thereof may be threaded, and the inner end **26A** of the rod **26** may be screwed into the end of the actuator of the air cylinder **30**. The outer end **26B** of the rod **26** is joined to the fastener puller **22** as described above.

(26) The pneumatic cylinder (or “air cylinder”) **30** is used to move the rod **26** and provide force on the head of a fastener using the fastener puller **22** in order to remove the fastener. The air cylinder **30** can comprise any suitable type of air cylinder. In the embodiment shown in the drawings, the fastener removal tool **20** comprises a linear tie rod pneumatic cylinder. The air cylinder **30** may comprise a tube or barrel cylinder **60** with an end cap **62A** and **62B** on each end to form an air chamber **64**. The air cylinder **30** may comprise a piston **66**; a piston rod **68**; and tie rods **70** that are joined to the end caps to hold the end caps in place. FIG. **3** shows that the air cylinder **30** may have forward and reverse flow inlets **72A** and **72B**, respectively, that are formed in portions of the air cylinder such as in the end caps **62A** and **62B**. FIG. **4** shows that the air cylinder **30** may also comprise: an air flow control valve **74**, an air direction control device **76**, and a muffler **78**. FIG. **2** shows that the air cylinder **30** may comprise air hoses **80** that are joined to the end caps of the air cylinder at hose fittings **81**. The air cylinder **30** may further comprise a quick disconnect hose coupler **82** (shown in FIG. **4**) for connecting the air cylinder **30** to a source of compressed air. The air cylinder **30** can be any suitable commercially available air cylinder. One suitable commercially available air cylinder is the MCMMASTER-CARR® square face compact air cylinder item 6113K232 available from McMaster-Carr Supply Company, Elmhurst, Illinois, U.S.A. The air

cylinder **30** is in operatively connected to the rod **26** as described above.

(27) The air flow control valve **74** controls the speed of air-powered equipment by adjusting the volume of air flow. The air flow control valve **74** can be any suitable commercially available air flow control valve. One suitable commercially available air flow control valve is item 62005K233 available from McMaster-Carr Supply Company. In some cases, the air flow control valve **74** can be replaced with another muffler **78**, such as McMaster-Carr high flow muffler item number 9835K42.

(28) The air flow direction control device (or “air manifold”) **76** is used to extend and then retract the piston in the air cylinder **30**. The air flow direction control device **76** can be any suitable commercially available air flow direction control device. One suitable commercially available air flow direction control device is the MCMaster-CARR® air directional control valve item 6859K51 available from McMaster-Carr Supply Company. This air flow direction control device **76** allows the piston rod **68** in the air cylinder **30** to be extended and retracted at different speeds. This device allows each action to be controlled by joining a flow control valve to each exhaust port. This particular device is known as a 4-way or 5/2 valve. The air flow direction control device **76** is provided with a push button actuator **77** that enables the operator to retract or extend the piston rod **68** of the air cylinder **30**. Other commercially available air flow direction control devices may be provided with a quick alternating solenoid that will cause the piston rod **68** of the pneumatic cylinder **30** to rapidly extend and retract. One such commercially available fast-switching, two-speed, two-action electrical air flow direction control device is MCMaster-CARR® item 2974N11. The air flow direction control device **76** can be joined to any suitable portion of the fastener removal tool **20**. In the embodiment shown, the air flow direction control device **76** is joined to a portion of the handle plate.

(29) The handle **34** is an optional, but useful feature of the fastener removal tool **20**. The handle **34**, in the embodiment shown, is part of a handle plate assembly **84** that is shown in detail in FIG. 4. In the embodiment shown, the handle plate assembly **84** comprises a handle plate and the handle **34**. The handle plate may be in the form of a plate **86** having a 90° bend in the same to form a first portion **86A** and a second portion **86B**. The handle **34** is attached to the first portion **86A** of the handle plate. The handle plate is used to join the handle **34** to the body of the fastener removal tool **20**. The handle **34** can be joined to any suitable portion of the fastener removal tool **20**. In the embodiment shown, the handle **34** is joined to the side of the air cylinder **30**. As shown in FIG. 2, the first portion **86A** of the handle plate is joined to the side of the air cylinder **30** with screws **88**. The handle **34** and handle plate assembly **84** can be made of any suitable material by any suitable process including those described for the stand-off mount **24**. In some cases, the handle **34** may comprise part of a 3D printed carbon fiber nylon combination receiving plate and handle.

(30) The fastener removal tool **20** may have additional or optional components. The optional or additional components can comprise separate inventive subject matter on a fastener removal tool **20** that does not have certain other features described herein (such as the ability to exert a pulling force on the head of a fastener in the form of a repeating intermittent strikes on the underside of the head of a fastener with said fastener puller). For example, the fastener removal tool **20** may comprise components for lubricating fasteners that are difficult to remove. For example, as shown in FIGS. 5-6 a liquid tank **90** may be joined to the pneumatic cylinder **30** with mounting brackets **92**. The liquid tank **90** may contain any suitable lubricating liquid including, but not limited to at least one of: a solvent, a penetrating lubricant, or liquid nitrogen. A liquid flow valve **94** may be joined to the end of the liquid tank **90** to dispense the desired amount of liquid. A liquid transport tube **96** connects to the end of the valve **94** and passes through a hole in the stand-off mount **24**. FIG. 7 shows the liquid transport tube **96** carries the liquid into a channel in the fastener puller **22** that dispenses the liquid onto the fastener.

(31) FIGS. 7A and 8 show that the fastener removal tool **20** may also comprise a system **100** for ensuring that the fastener puller **22** is properly positioned to remove a fastener. For example, a

sensor **102** can be used to determine if the fastener puller **22** is correctly positioned to remove a fastener. The sensor **102** can be any suitable type of sensor including, but not limited to one or more of the following: an optical sensor, a pressure sensor, an electrical sensor, a laser sensor, a strain sensor, and a proximity switch. The sensor **102** can be placed in any suitable location. In some cases, the sensor **102** may be placed in the recessed pocket **38** of the fastener puller **22** to sense when the fastener puller **22** is correctly positioned around the head of a fastener. In one example, an electrical sensor can be provided and positioned so that the fastener will complete a circuit. In the example shown in the drawings, the sensor **102** comprises a flexible material that has an electrical contact on it that completes a circuit when it is pressed in by the fastener being pushed against the back of the fastener puller **22**.

(32) The components of the fastener removal tool **20** can be made in any suitable manner, and from any suitable materials. The standoff mount **24**, stand-off mount-receiving component **46**, and handle plate assembly **84** may, for example, be 3D printed on a Stratasys FORTUS™ 450 MC 3D printer, available from Stratasys, Ltd. of Eden Prairie, MN, U.S.A. in nylon 12 carbon fiber. The air manifold/flow direction control device **76**, pneumatic alternating solenoid, quick disconnect air fittings, air hoses **80**, air flow mufflers, nuts, bolts, washers, and threaded rod **26**, and air cylinder **30** are off the shelf items that are connected to the handle plate assembly **84**, stand-off mount receiving component **46**, and standoff mount **24** using the nuts and bolts. It is expressly not admitted, however, that any of the components are known for being assembled into a fastener removal tool **20** as described herein.

(33) The components of the fastener removal tool **20** can be arranged in any suitable manner. In the embodiment shown, the push button actuator **77**, the pneumatic cylinder **30**, standoff mount **24**, threaded rod **26**, and fastener puller **22** may be assembled in line with each other and act in a linear motion that is aligned with the target fastener to be removed. The handle **34** protrudes out perpendicular to the body of the rest of the device **20**. In this configuration, the operator operates the tool with both hands simultaneously, thus providing a degree of safety by keeping the operator's hands away from any powered components. The quick disconnect air fitting **82** is attached to the inlet of the air manifold **76**, the two exhaust mufflers **78** to the exits of the manifold, and the air hoses **80** from the controlled exits of the manifold to the inlets of the air cylinder.

(34) The components of the fastener removal tool **20** can be assembled as follows. The fastener puller **22** is joined to the tie rod cylinder **30** using the threaded rod **26** and locked in place using a set screw **36**. The stand-off mount **24** is slipped onto the standoff mount receiving component **46** and held in place using a set screw **36**. The quick disconnect fittings, and air flow mufflers **78** are screwed into the air flow regulator/flow direction control device **76**. The air hoses **80** connect the differing flow exits from the control device to the forward and reverse flow inlets on the tie rod cylinder **30**. The liquid tank **90** may be joined to the pneumatic cylinder **30** with mounting brackets **92**. The liquid flow valve **94** is joined to the end of the liquid tank **90** and the liquid transport tube **96** connects to the end of the valve and carries the liquid into a channel in the fastener puller **22**. The pressure sensor is placed in the back recessed pocket **38** of the fastener puller **22** and connected to an indicator such as a light **104** (FIG. 6). The indicator can be any suitable type of indicator. The indicator can be in any suitable location. In the embodiment shown, the light **104** is situated near the push button mechanism **77**.

(35) The fastener removal tool **20** can be used as shown in FIGS. 9 and 10. The tool **20** is used for the removal of fasteners **12** that become stuck after the bottom of the fastener head **14** is exposed. The fastener may have been partially removed through various means and then became stuck with a portion of the neck exposed above the plane of the work surface. The fastener removal tool **20** will function on any fastener **12** that is stuck in a surface **10** (or in its housing that is embedded in a surface) as long as the bottom of the fastener head **14** is exposed and the stand-off mount **24** can brace against the surface **10** of the article containing the fastener. Thus, it may be desirable for the work surface to be flat enough, at least in the area of the fastener, for the standoff mount to (have

sufficient purchase to) rest against.

(36) In the case of a pneumatic fastener removal tool **20**, the fastener removal tool **20** is connected to a source of compressed air. The compressed air causes the piston rod **68** (shown in FIG. **3**) of the pneumatic cylinder **30** to extend and the puller **22** to protrude slightly from the standoff mount **24** (as shown in FIG. **1**). The fastener puller **22** is slid around the neck **16** and under the head **14** of the fastener **12**. If the fastener removal tool **20** is provided with a sensor system, this activates the pressure sensor **102** at the back of the fastener puller **22**. This may illuminate a light **104** that is positioned near the actuator, such as actuation button **77** that indicates the fastener is correctly placed in the fastener puller **22**. The operator then presses and holds the actuation button **77** on the flow control device. This causes the air flow direction to change, and the cylinder **30** retracts slowly which pulls the fastener out. The operator then releases the button, and the cylinder **30** returns to an extended position where the fastener can now be slid out of the puller **22** and removed therefrom. Should the fastener fail to extract, the operator can release the lubricating liquid from the liquid tank **90** to assist in breaking the fastener loose.

(37) FIG. **10** shows that the operator also has the option to activate the quick alternating solenoid that will cause the piston rod **68** of the pneumatic cylinder **30** to rapidly extend and retract as it pulls up on the fastener head **14**. FIG. **11** shows that the fastener removal tool **20** may be provided with a selector, such as a switch **110**, for selecting the type of movement of the first rod between two modes of applying a force to remove a fastener: (a) exerting a continuous pulling force with said fastener puller, or (b) exerting a pulling force in the form of repeating intermittent strikes on the underside of the head of a fastener with the fastener puller. FIGS. **9** and **10** show the exertion of a pulling force in the form of a repeating intermittent strikes on the underside of the head of a fastener schematically. FIG. **9** shows the fastener puller **22** in a first position, position number 1. FIG. **10** shows the fastener puller **22** in positions numbered 2-5. In this operation, the fastener puller **22** loses contact with the bottom of the fastener head **14** while it extends and strikes it with increased force when the direction switches and the piston rod **68** retracts, hammering the fastener head **14** from underneath in an effort to extract the fastener.

(38) There are numerous, non-limiting embodiments of the invention. All embodiments, even if they are only described as being “embodiments” of the invention, are intended to be non-limiting (that is, there may be other embodiments in addition to these), unless they are expressly described as limiting the scope of the invention. Any of the embodiments described herein can also be combined with any other embodiments in any manner to form still other embodiments.

(39) The fastener puller **22** and the standoff mount **24** can be easily removed and replaced with differently sized pieces in order to fit various fasteners and surface situations. The air flow control manifold **76** can be changed to similarly functioning air flow control devices as long as they include a 2-direction air flow and exhaust capability. The pneumatic tie rod cylinder **30** can be replaced with a similarly sized and similarly powerful, or stronger, 2-way actuating pneumatic cylinder. A trigger can be added to the handle **34** in order to control air flow direction rather than the push button design shown. An oscillation system can be added to the air flow control to cause the air cylinder to rapidly oscillate in and out as it retracts and pulls the fastener out. An air pressure regulator can be joined to the fastener removal tool **20** in order to control the force exerted by the pneumatic cylinder.

(40) In other embodiments, the pneumatic cylinder **30**, air flow control manifold **76**, air hoses **80**, and other components of the pneumatic system can be replaced with components of a hydraulic system. In such a case, rather than being connected to a supply of compressed air, the hydraulic system will be connected to a supply of hydraulic fluid. In still other embodiments, the pneumatic system components may be replaced with components of a magnetic, electric, or electronic system. In these latter embodiments, an electric motor can be used to power the system.

(41) The fastener removal tool **20** described herein can provide a number of advantages. It should be understood, however, that these advantages need not be required unless they are set forth in the

appended claims.

(42) The fastener removal tool **20** greatly improves the efficiency and safety of the process to remove stuck fasteners. This tool will relieve the user of any manual labor and increase their ability to remove difficult to extract fasteners. The fastener removal tool **20** will also protect the surface of the item to which the fasteners are connected. The fastener removal tool **20** will protect the user of the tool from debris, and will actuate in a smooth controllable motion. The configuration of the fastener removal tool ensures that the operator places both hands on the tool to reduce the possibility of injury to the operator's appendages. The fastener removal tool **20** will be useful for multiple differently sized fasteners, and will be useful in a wide range of conditions and spaces. (The tool itself employs a pneumatic cylinder to extract the fasteners in a linear motion.)

(43) As used herein, an element or step recited in the singular and preceded with the word “a” or “an” should be understood as including the plural of such elements or steps, unless the plural of such elements or steps is specifically excluded.

(44) The term “joined”, as used herein, encompasses configurations in which an element is directly secured to another element by affixing the element directly to the other element; configurations in which the element is indirectly secured to the other element by affixing the element to intermediate member(s) which in turn are affixed to the other element; and configurations in which one element is integral with another element, i.e., one element is essentially part of the other element. The term “joined” includes both those configurations in which an element is temporarily joined to another element, or in which an element is permanently joined to another element.

(45) While the present invention has been illustrated by a description of one or more embodiments thereof and while these embodiments have been described in considerable detail, they are not intended to restrict or in any way limit the scope of the appended claims to such detail. Additional advantages and modifications will readily appear to those skilled in the art. The invention in its broader aspects is therefore not limited to the specific details, representative apparatus and method, and illustrative examples shown and described. Accordingly, departures may be made from such details without departing from the scope of the general inventive concept.

Claims

1. A fastener removal tool for removing a fastener having a head that is partially protruding from the surface of an article, said fastener removal tool comprising: a fastener puller for engaging the head of a fastener, the head of the fastener having a top surface and an underside; a stand-off mount for placing against a surface having a fastener therein, said stand-off mount having two ends comprising a surface-contacting end and an opposing end, and a passageway between said ends; a first rod joined to said fastener puller and passing at least part of the way through the passageway in the stand-off mount; a power source having a piston rod joined to said first rod, wherein the power source controls the motion of said first rod; and a selector in communication with the power source for selecting the type of movement of the first rod between two modes of applying a force to remove a fastener: (a) exerting a continuous pulling force with said fastener puller, or (b) exerting a pulling force in the form of repeating intermittent strikes on the underside of the head of a fastener with said fastener puller.

2. A fastener removal tool for removing a fastener having a head that is partially protruding from the surface of an article, said fastener removal tool comprising: a fastener puller for engaging the head of a fastener, the head of the fastener having a top surface and an underside; a stand-off mount for placing against a surface having a fastener therein, said stand-off mount having two ends comprising a surface-contacting end and an opposing end, and a passageway between said ends; a first rod joined to said fastener puller and passing at least part of the way through the passageway in the stand-off mount; a power source having a piston rod joined to said first rod, wherein the power source controls the motion of said first rod; and a sensor for determining if the fastener

puller is properly positioned to remove a fastener, wherein said fastener removal tool is configured to have one setting in which the fastener removal tool exerts a pulling force on the head of a fastener in the form of repeating intermittent strikes on the underside of the head of a fastener with said fastener puller.

3. A fastener removal tool for removing a fastener having a head that is partially protruding from the surface of an article, said fastener removal tool comprising: a fastener puller for engaging the head of a fastener, the head of the fastener having a top surface and an underside; a stand-off mount for placing against a surface having a fastener therein, said stand-off mount having two ends comprising a surface-contacting end and an opposing end, and a passageway between said ends; a first rod joined to said fastener puller and passing at least part of the way through the passageway in the stand-off mount; a power source having a piston rod joined to said first rod, wherein the power source controls the motion of said first rod; and an indicator mechanism for indicating whether the fastener puller completely engages a fastener, wherein the indicator mechanism comprises a pressure sensor inside of the fastener puller that illuminates a light to indicate that the fastener puller is correctly placed on the head of the fastener, wherein said fastener removal tool is configured to have one setting in which the fastener removal tool exerts a pulling force on the head of a fastener in the form of repeating intermittent strikes on the underside of the head of a fastener with said fastener puller.

4. A fastener removal tool for removing a fastener having a head that is partially protruding from the surface of an article, said fastener removal tool comprising: a fastener puller for engaging the head of a fastener, the head of the fastener having a top surface and an underside; a stand-off mount for placing against a surface having a fastener therein, said stand-off mount having two ends comprising a surface-contacting end and an opposing end, and a passageway between said ends; a first rod joined to said fastener puller and passing at least part of the way through the passageway in the stand-off mount; a power source having a piston rod joined to said first rod, wherein the power source controls the motion of said first rod; and a fastener lubricating system joined to the fastener removal tool, wherein said fastener lubricating system is configured to deliver a lubricating liquid onto the fastener being removed, wherein said fastener removal tool is configured to have one setting in which the fastener removal tool exerts a pulling force on the head of a fastener in the form of repeating intermittent strikes on the underside of the head of a fastener with said fastener puller.

5. A fastener removal tool for removing a fastener having a head that is partially protruding from the surface of an article, said fastener removal tool comprising: a fastener puller for engaging the head of a fastener, the head of the fastener having a top surface and an underside; a stand-off mount for placing against a surface having a fastener therein, said stand-off mount having two ends comprising a surface-contacting end and an opposing end, and a passageway between said ends; a first rod joined to said fastener puller and passing at least part of the way through the passageway in the stand-off mount; a power source having a piston rod joined to said first rod, wherein the power source controls the motion of said first rod; and a fastener lubricating system joined to the fastener removal tool, wherein the fastener lubricating system comprises a liquid tank containing a lubricating liquid that is joined to said pneumatic cylinder, and a liquid transport tube connects to the end of the valve and carries the liquid into a channel in the fastener puller, wherein said fastener removal tool is configured to have one setting in which the fastener removal tool exerts a pulling force on the head of a fastener in the form of repeating intermittent strikes on the underside of the head of a fastener with said fastener puller.

6. The fastener removal tool of claim 5 wherein the lubricating liquid comprises at least one of: a solvent, a penetrating lubricant, and liquid nitrogen.

7. A fastener removal tool for removing a fastener having a head that is partially protruding from the surface of an article, said fastener removal tool comprising: a fastener puller for engaging the head of a fastener, the head of the fastener having a top surface and an underside; a stand-off mount

for placing against a surface having a fastener therein, said stand-off mount having two ends comprising a surface-contacting end and an opposing end, and a passageway between said ends; a first rod joined to said fastener puller and passing at least part of the way through the passageway in the stand-off mount; a power source having a piston rod joined to said first rod, wherein the power source controls the motion of said first rod; and wherein said fastener removal tool is configured to have one setting in which the fastener removal tool exerts a pulling force on the head of a fastener in the form of repeating intermittent strikes on the underside of the head of a fastener with said fastener puller, wherein the piston rod extends and retracts when the fastener removal tool exerts a pulling force on the head of a fastener in the form of repeating intermittent strikes on the underside of the head of a fastener with said fastener puller, wherein when the piston rod extends, the fastener puller loses contact with the underside of the head of the fastener, and when the piston rod retracts, the fastener puller strikes the underside of the fastener head with increased force, hammering the fastener head from underneath.

8. A method of removing a fastener from an article, the fastener having a body and a head joined to the body, wherein the fastener head has a top surface and an underside, wherein the fastener head protrudes from the surface of the article so that a gap is provided between the underside of the fastener head and the surface of the article, said method comprising: a) providing a powered fastener removal tool comprising: a fastener puller for engaging the head of a fastener; a stand-off mount for placing against a surface of an article having a fastener therein, wherein the stand-off mount has a passageway therethrough and the fastener puller is positioned for reciprocating motion in said passageway; and a power source operatively connected to said fastener puller, wherein the power source controls the motion of said fastener puller; b) placing the stand-off mount against the surface of the article and engaging the head of the fastener with the fastener puller; and c) exerting a repeating intermittent striking force on the underside of the head of a fastener with the fastener puller using said powered fastener removal tool, wherein the reciprocating motion of the fastener puller is such that in step e), when the fastener removal tool exerts a pulling force on the head of a fastener in the form of repeating intermittent strikes on the underside of the head of a fastener with said fastener puller, the fastener puller loses contact with the underside of the head of the fastener, and the fastener puller then switches directions and the fastener puller strikes the underside of the fastener head with increased force, hammering the fastener head from underneath.

9. A fastener removal tool for removing a fastener having a head that is partially protruding from the surface of an article, said fastener removal tool comprising: a fastener puller for engaging the head of a fastener, the head of the fastener having a top surface and an underside; a stand-off mount for placing against a surface having a fastener therein, said stand-off mount having two ends comprising a surface-contacting end and an opposing end, and a passageway between said ends; a first rod joined to said fastener puller and passing at least part of the way through the passageway in the stand-off mount; a power source having a piston rod joined to said first rod, wherein the power source controls the motion of said first rod, wherein the power source is a pneumatic cylinder; and an air flow direction control device operationally connected to the pneumatic cylinder, wherein the air flow direction control device comprises an alternating solenoid, wherein said fastener removal tool is configured to have one setting in which the fastener removal tool exerts a pulling force on the head of a fastener in the form of repeating intermittent strikes on the underside of the head of a fastener with said fastener puller, wherein the alternating solenoid causes the piston rod to extend and retract which provides said repeating intermittent strikes.

10. A method of removing a fastener from an article, the fastener having a body and a head joined to the body, wherein the fastener head has a top surface and an underside, wherein the fastener head protrudes from the surface of the article so that a gap is provided between the underside of the fastener head and the surface of the article, said method comprising: a) providing a powered fastener removal tool comprising: a fastener puller for engaging the head of a fastener; a stand-off mount for placing against a surface of an article having a fastener therein, wherein the stand-off

mount has a passageway therethrough and the fastener puller is positioned for reciprocating motion in said passageway; a power source operatively connected to said fastener puller, wherein the power source controls the motion of said fastener puller, wherein the power source is a pneumatic cylinder; and an air flow direction control device operationally connected to the pneumatic cylinder, wherein the air flow direction control device comprises an alternating solenoid that causes the fastener puller to extend and retract; b) placing the stand-off mount against the surface of the article and engaging the head of the fastener with the fastener puller; and c) exerting a repeating intermittent striking force on the underside of the head of a fastener with the fastener puller using said powered fastener removal tool.
